

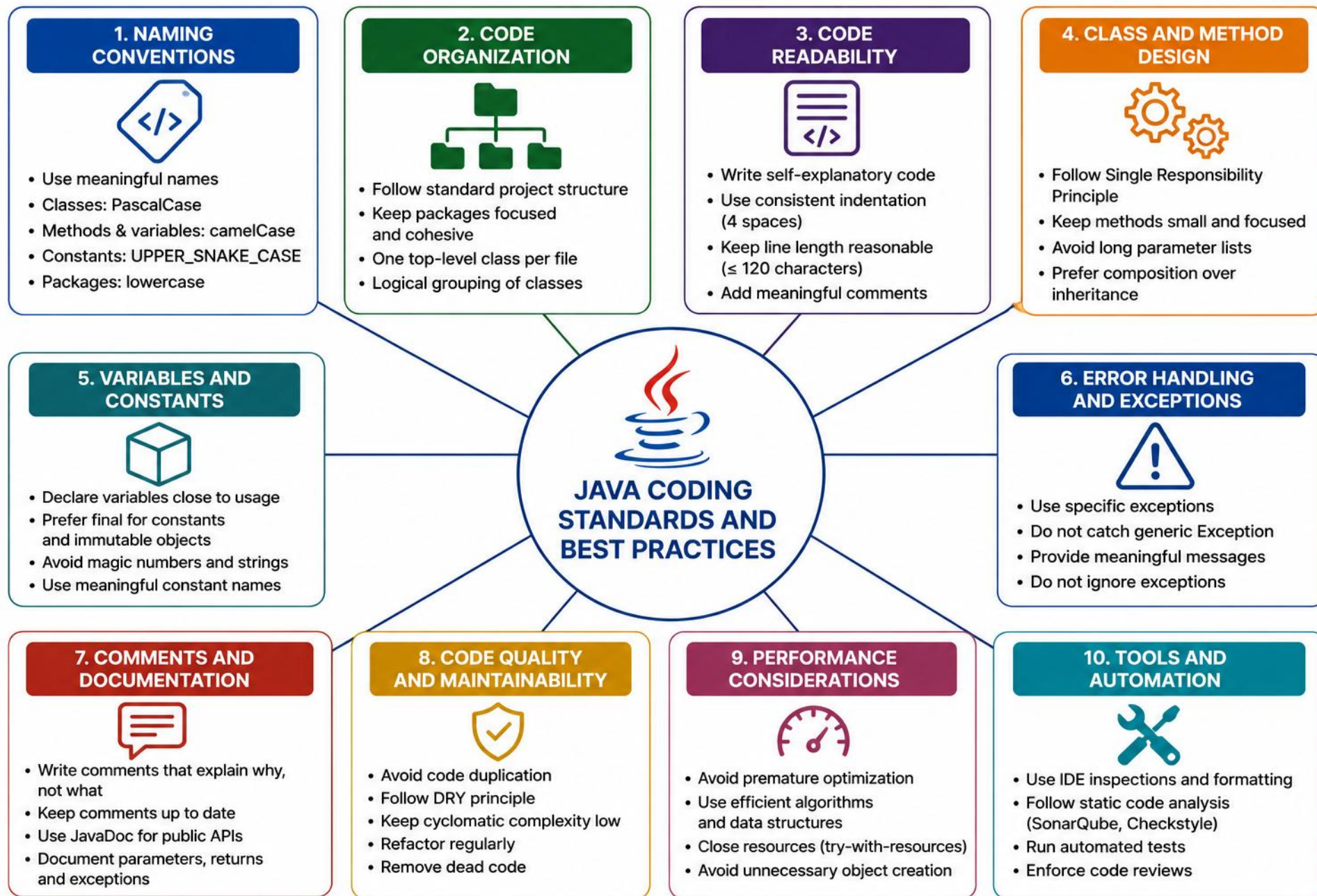
MODULE 12

Java Coding Standards and Best Practices

Write clean, consistent, maintainable Java code that's easy to read, test, and evolve in enterprise environments.







MODULE 12 OVERVIEW

Enterprise systems live for years — code must be readable, maintainable, and consistent.

- Building on Modules 10-11's AI-assisted workflows, this module covers the coding standards and refactoring discipline that keeps a codebase healthy long after AI generates the first draft.

DURATION & LAB



2 Hours Theory

Standards, design principles, refactoring.



45 Minutes Lab

Coding Standards and Refactoring.



Scenario

Bring a legacy CRM CustomerService up to standard.

STANDARDS COVER

Naming, formatting, structure, testability, docs

TOOLS ENFORCE

Checkstyle, PMD, SpotBugs, SonarQube, IDE inspections

THE PAYOFF

Fewer defects, faster onboarding, lower cost

Learning objectives: apply naming/formatting conventions, design clean classes and methods, handle exceptions and logging, write testable code.

KEY TAKEAWAY Consistent standards + clean code + regular reviews = high-quality, maintainable Java applications.

WHY CODING STANDARDS MATTER

Coding standards are a shared set of rules and conventions — the foundation of professional, maintainable software.

WITHOUT STANDARDS

- Inconsistent code — hard to read and understand
- Difficult collaboration and onboarding
- Harder maintenance, more defects
- Slower delivery, higher cost

VS

WITH STANDARDS

- Consistency and predictability across the codebase
- Stronger collaboration and shared understanding
- Better quality, easier long-term maintenance
- Faster delivery, lower risk and cost

THE BUSINESS IMPACT



Lower Risk



Cost Savings



Faster Time to Market



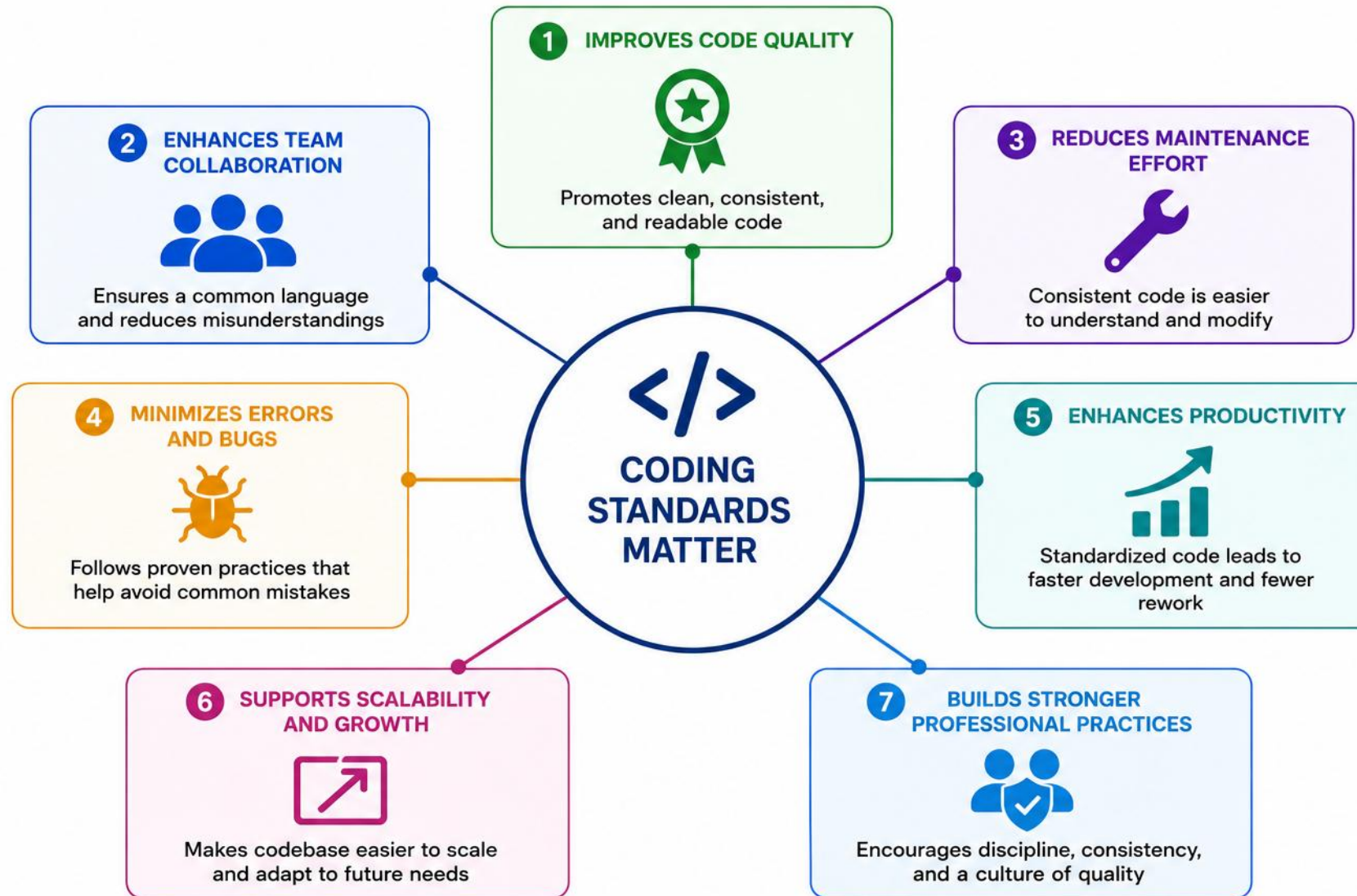
Scalable Growth



Professional Excellence

KEY TAKEAWAY Coding standards aren't about restrictions — they enable clarity, consistency, and quality.

Why Coding Standards Matter



ENTERPRISE CODING STANDARDS

Enterprise standards define how we write, structure, and document code across the organization.

CORE STANDARDS IN THIS MODULE



Naming

- Consistent, meaningful identifiers.



Formatting

- Automated indentation and style.



Method & Class Design

- Cohesive, well-structured units.



Readability & Complexity

- Clear code; complexity kept manageable.



Refactoring

- Safe, incremental improvement.



Static Analysis & Code Review

- Automated tooling plus team review.

CROSS-REFERENCES

- Exceptions — covered in depth in Module 7.
- Structure — covered in depth in Module 8.
- Dependencies — covered in depth in Module 9.
- Security — covered in a later module.

KEY TAKEAWAY Enterprise coding standards create a common language, improve quality, reduce risk, and accelerate delivery.



NAMING CONVENTIONS

Use meaningful names. Write self-documenting code — consistent naming makes code easier to read and maintain.

ELEMENT	STYLE	RULE	EXAMPLE
Package	lowercase	Reverse domain name	com.company.project
Class / Interface	PascalCase	Noun or noun phrase	OrderService
Method	camelCase	Verb or verb phrase	calculateTotalAmount()
Variable	camelCase	Meaningful, no encodings	customerCount
Constant (static final)	UPPER_SNAKE_CASE	Words separated by _	MAX_RETRY_COUNT
Enum type	PascalCase	Noun, singular	PaymentStatus
Enum constant	UPPER_SNAKE_CASE	Words separated by _	PAYMENT_COMPLETED
Type Parameter	Single letter	Letter, or descriptive if unclear	<T>, <K, V>, <TEntity>
Field (instance)	camelCase	Noun or noun phrase	orderRepository

BEST PRACTICES

**Be Descriptive**

**Be Consistent**

**Concise, Not Short**

**No Reserved Words**

KEY TAKEAWAY Good names are the foundation of clean code — follow conventions, be consistent, make your code self-documenting.

Naming Conventions

1 CLASSES



- Use PascalCase
- Noun or noun phrase
- Singular

Example:
`CustomerService`

2 INTERFACES



- Use PascalCase
- Adjective or noun phrase
- Often start with an "I"

Example:
`IPaymentProcessor`

3 METHODS



- Use camelCase
- Verb or verb phrase
- Start with lowercase letter

Example:
`calculateTotal()`

4 VARIABLES



- Use camelCase
- Noun or noun phrase
- Start with lowercase letter

Example:
`customerName`

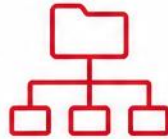
5 CONSTANTS



- Use UPPER_SNAKE_CASE
- Noun or noun phrase
- Use meaningful names

Example:
`MAX_RETRY_COUNT`

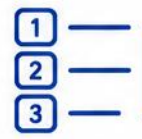
6 PACKAGES



- Use lowercase
- Use nouns
- Separate words with dots

Example:
`com.company.project.service`

7 ENUMS



- Use PascalCase
- Singular noun
- Group related constants

Example:
`OrderStatus`

8 TYPE PARAMETERS



- Use single uppercase letter
- Use meaningful letters
- Common: T, E, K, V

Example:
`Map<K, V>`

METHOD DESIGN PRINCIPLES AND CHECKLIST

Well-designed methods make code easy to read, test, and maintain — with a quick checklist and naming examples to apply them.



Single Responsibility

- A method should do one thing and do it well.



Short and Focused

- Easy to follow at one level of abstraction — length is a signal, not a rule.



Clear Parameters

- Related values traveling together? Consider a parameter object.



Avoid Side Effects

- Don't modify external state unexpectedly.



Return Meaningful Values

- Return a value when callers need one; void suits clear commands (e.g., `sendNotification()`).



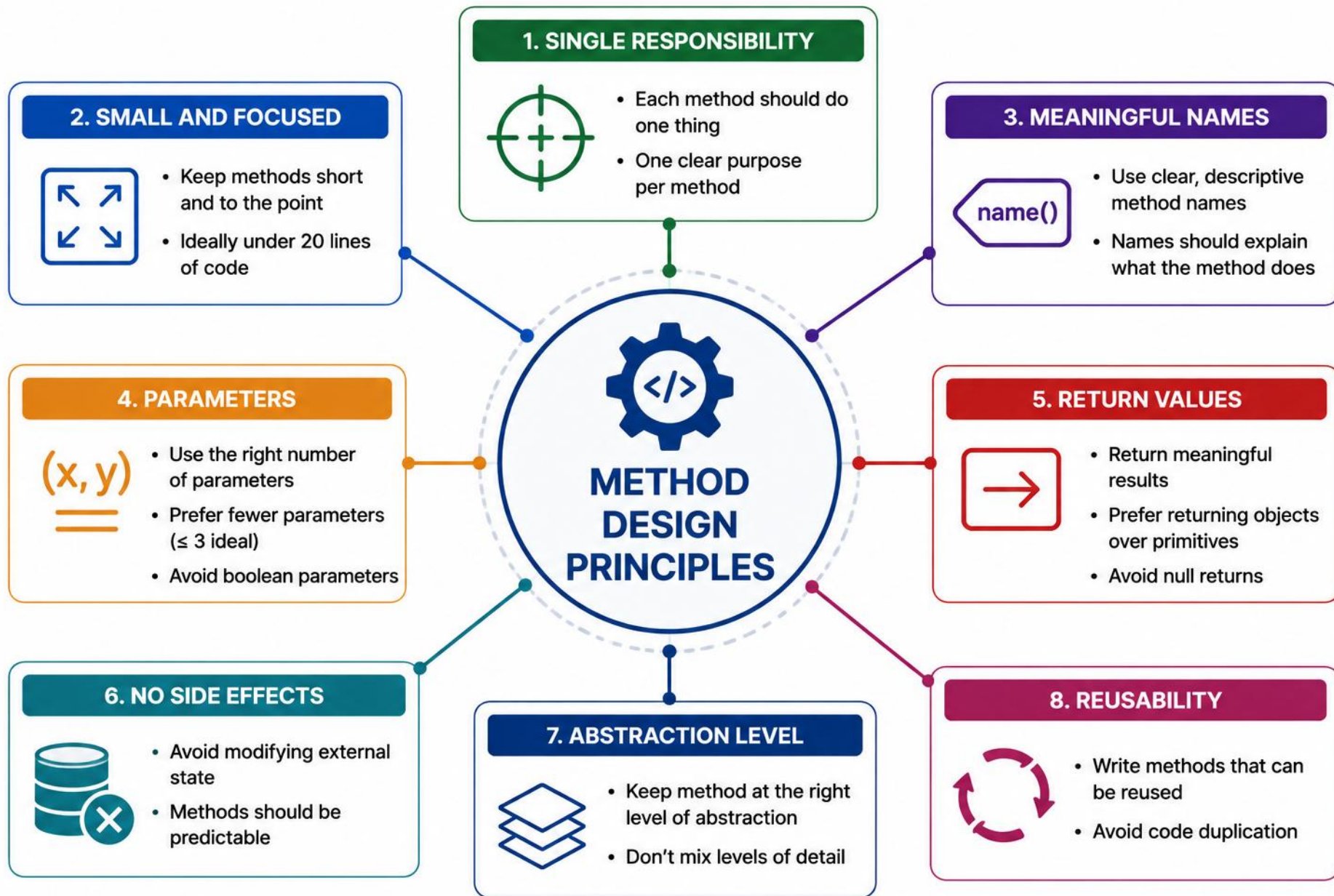
Handle Exceptions

- Don't swallow exceptions; add context.

ANTI-PATTERNS TO AVOID

- Long methods mixing responsibilities; parameter lists that are hard to follow; generic names like `process()`, `handle()`, `doStuff()`.
- Good names describe purpose: `calculateInvoiceTotal()`, `isEligibleForDiscount()`, `validateEmailAddress()`.
- Avoid vague names: `process()`, `run()`, `handleData()`, `temp()`, `doStuff()`.
- Checklist: name matches purpose; minimal meaningful parameters; no duplication; guard clauses; easy to test in isolation.

KEY TAKEAWAY If you can't explain a method's purpose in one sentence, it's doing too much.



CLASS DESIGN PRINCIPLES: SOLID IN PRACTICE

Well-designed classes are the foundation of clean, maintainable software — with anti-patterns to avoid and quick tips for applying SOLID.



High Cohesion

- A class should represent one cohesive responsibility — not mix unrelated business, infrastructure, and presentation concerns.



Encapsulation

- Hide internal state with private fields.



Abstraction

- Expose only essential details through public methods.



Open/Closed Principle

- Design stable extension points where variation is expected; refactor existing code when that's simpler and clearer.



Favor Composition

- Use composition over inheritance for flexibility.



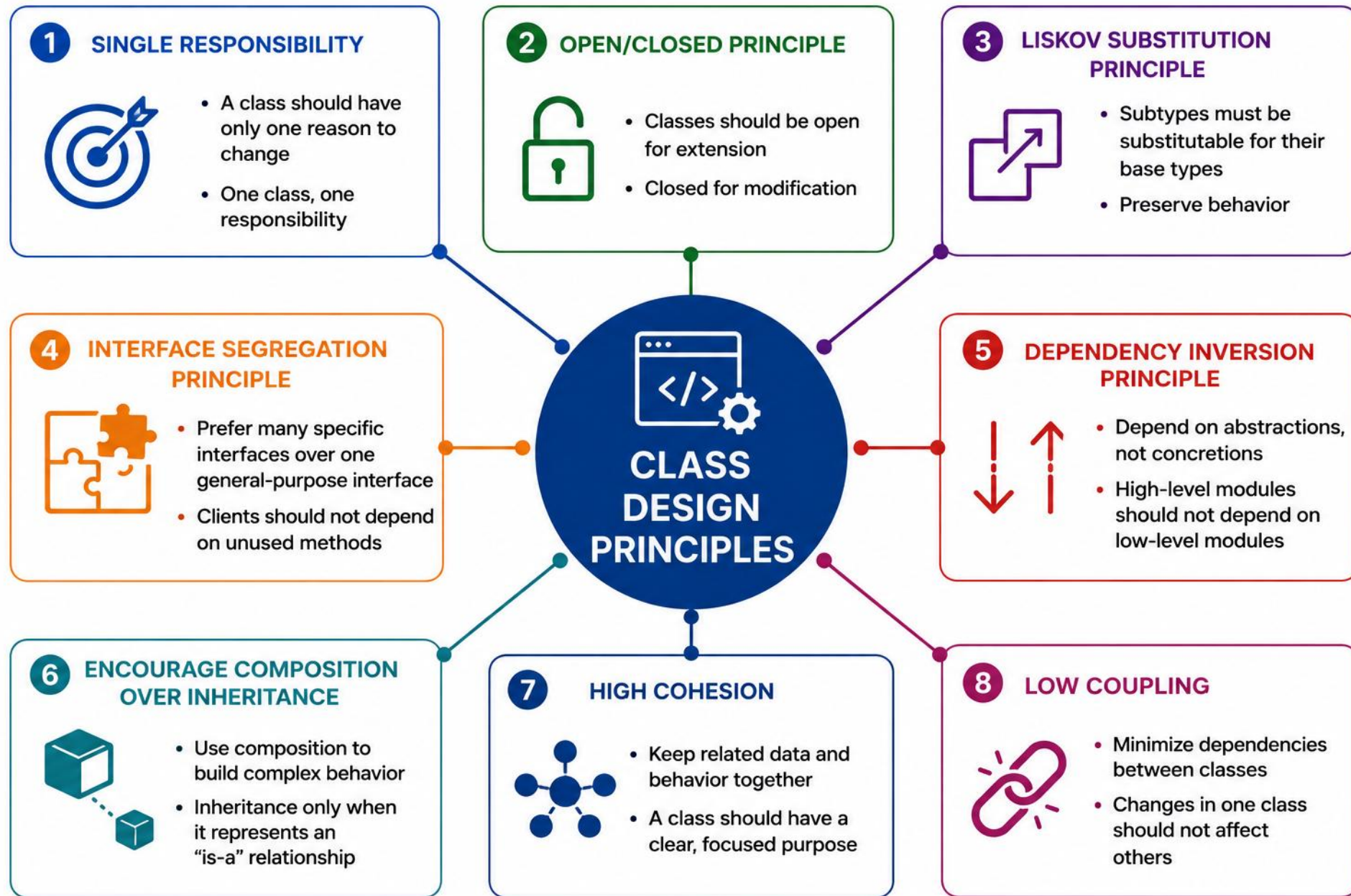
Keep Classes Small

- Small classes are easier to understand and test.

SOLID PRINCIPLES

- Single Responsibility • Open/Closed • Liskov Substitution • Interface Segregation • Dependency Inversion.
- ANTI-PATTERNS TO AVOID — God classes with too many responsibilities; public fields exposing internal state; inheritance used where composition fits better; deep hierarchies; unclear class names.

KEY TAKEAWAY If the class name is hard to explain, rethink the design — encapsulate data and keep classes focused.



BEFORE / AFTER: READABILITY REFACTOR

Readable code is easier to understand, review, test, and extend — see the difference a small refactor makes.

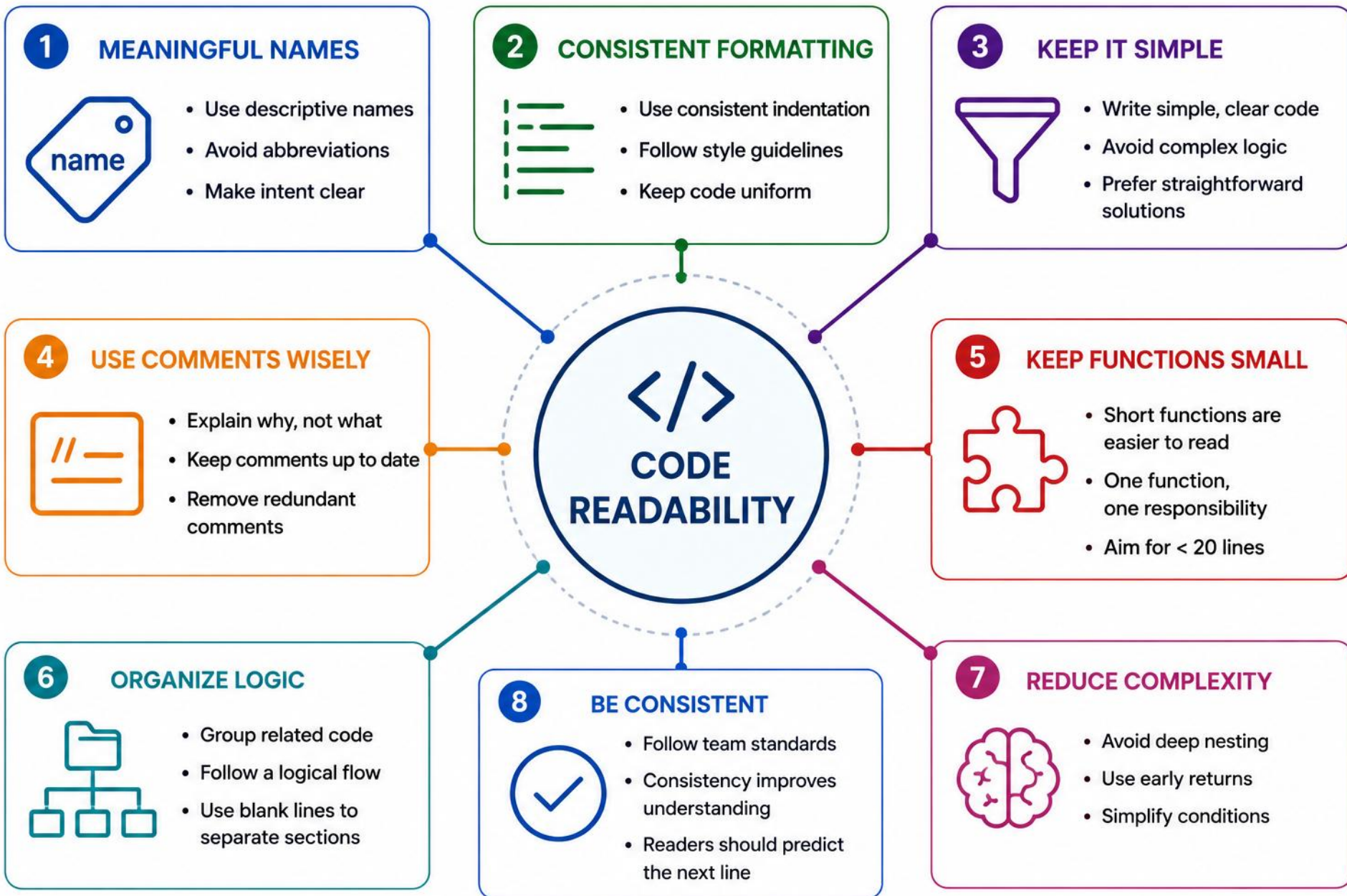
BEFORE / AFTER EXAMPLE

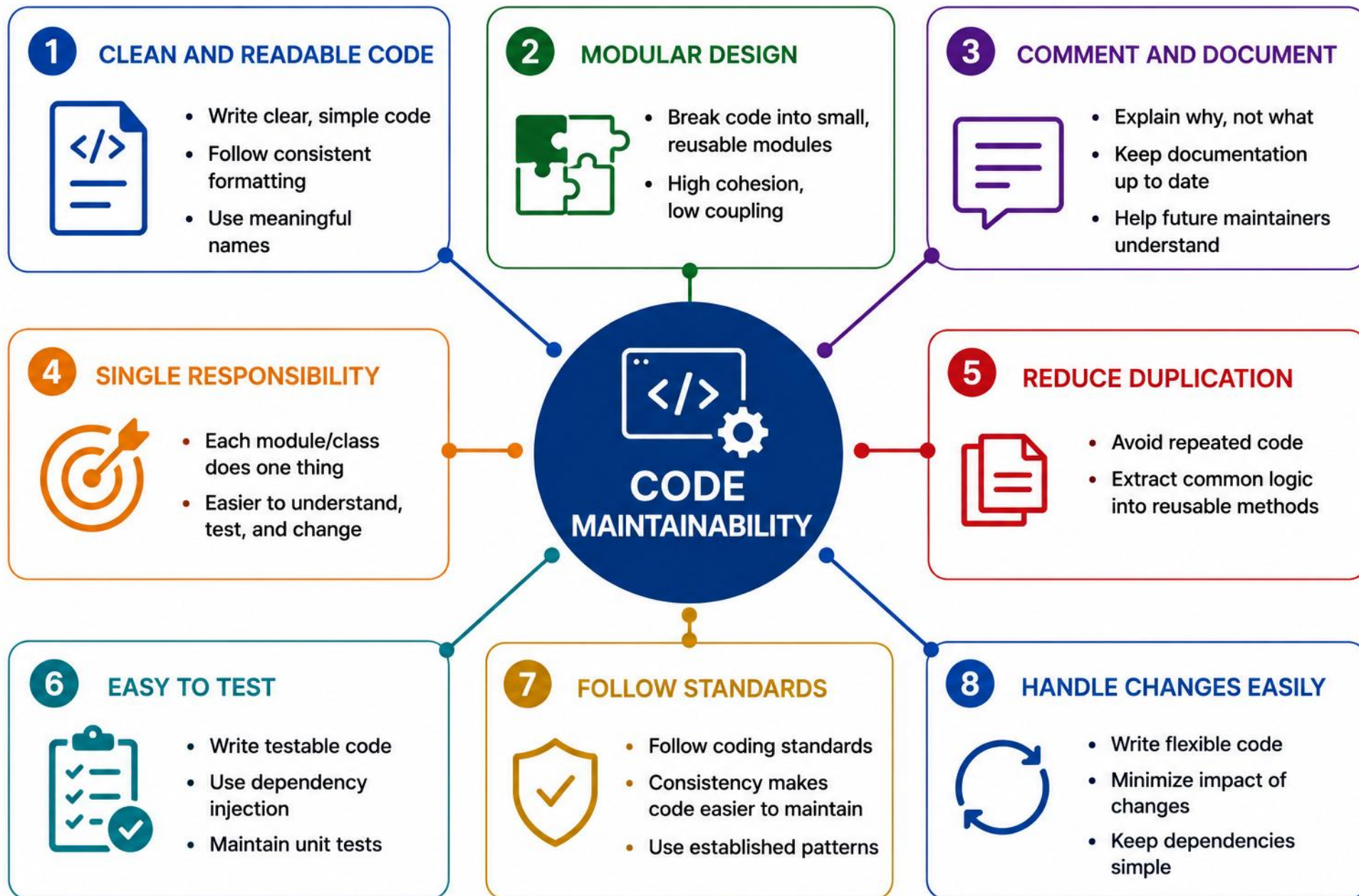
```
// Bad
int d; String n;

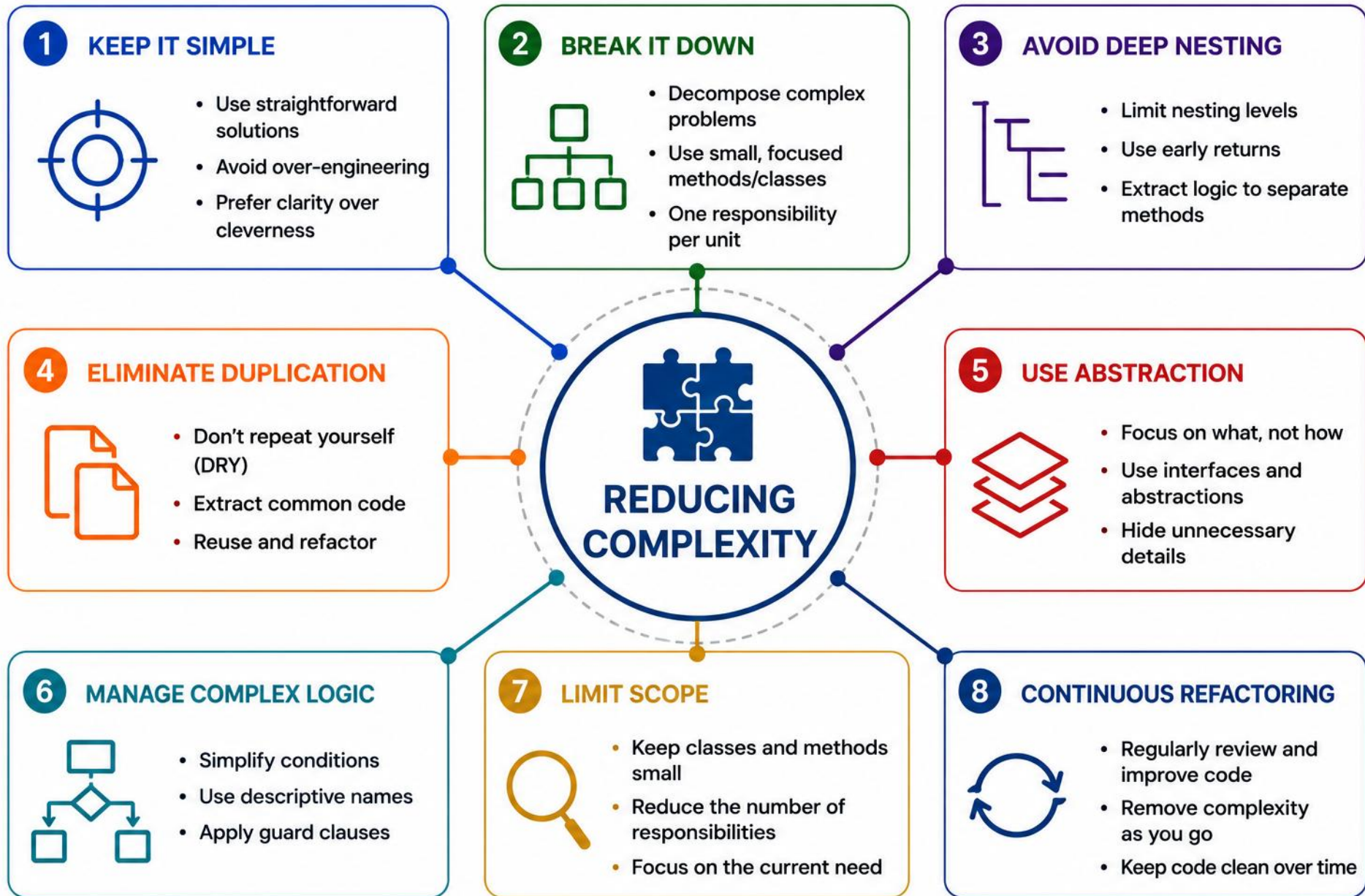
// Good
int daysSinceLastLogin;
String customerName;
```

- Quick tips: read your code aloud, use IDE formatting tools, refactor early and often.
- Tools that help: Checkstyle, PMD, SonarQube, IntelliJ IDEA.

KEY TAKEAWAY Clean, well-structured code today becomes easy to maintain, extend, and scale tomorrow.





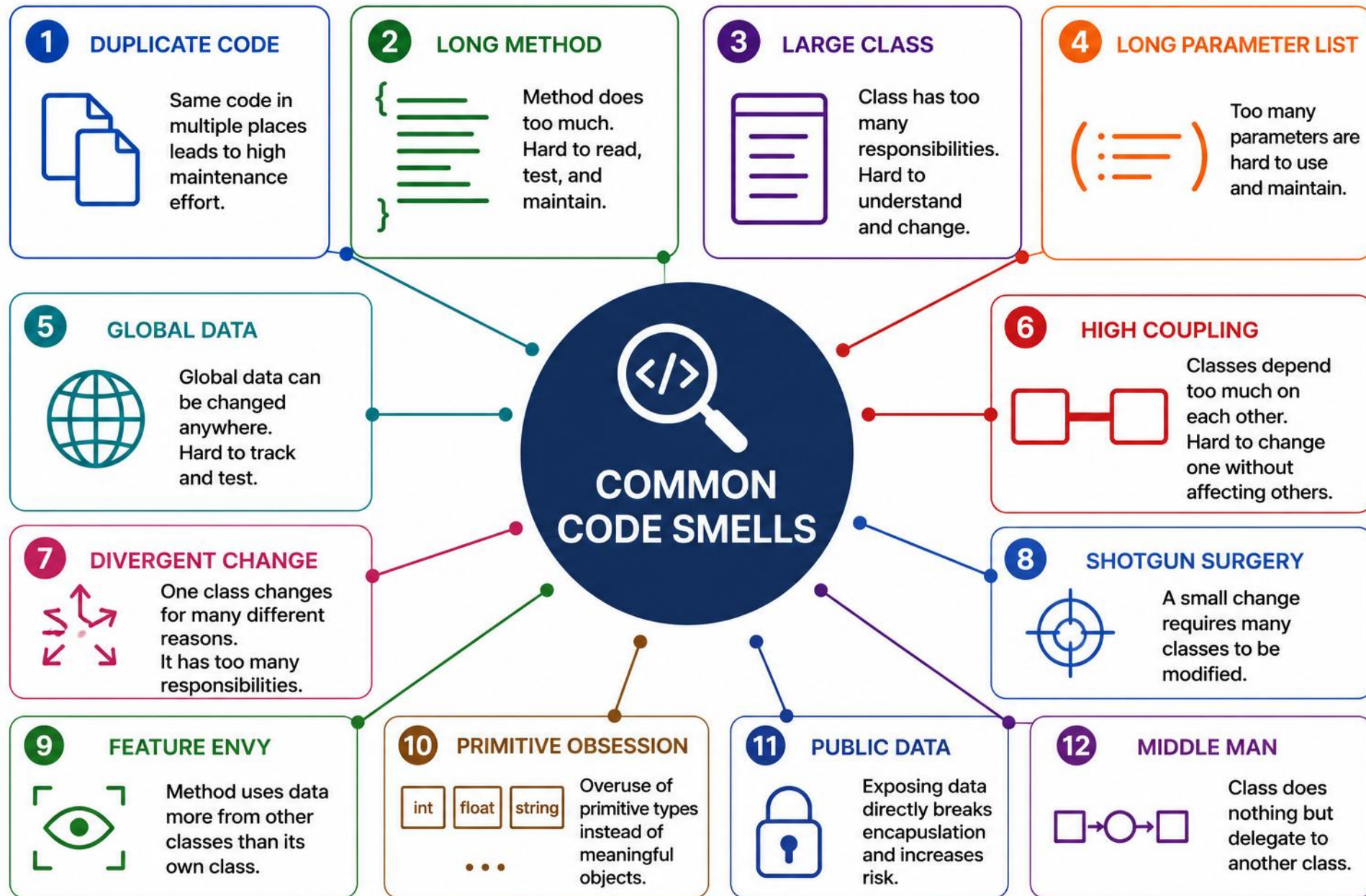


COMMON CODE SMELLS

Code smells don't always mean the code is wrong — they signal something may be harder to maintain or change.

CODE SMELL	WHAT IT LOOKS LIKE	HOW TO IMPROVE
Long Method	A method is very long, doing many things	Extract Method — break into small, focused methods
Duplicated Code	Same code appears in multiple places	Extract Method — create one method and reuse it
Long Parameter List	Method requires many parameters (4+)	Introduce Parameter Object
Deep Nesting	Multiple levels of if/else or loops	Guard Clauses — return early, reduce nesting
Magic Numbers/Strings	Hard-coded literals scattered in code	Use Constants/Enums
Shotgun Surgery	One small change requires edits across many classes	Centralize the rule in one class (e.g., DiscountPolicy)
God Class / Feature Envy	A class does too much, or a method overly depends on another class's data	Apply Single Responsibility; move logic to the class that owns the data
Data Clumps	The same group of variables or parameters travels together repeatedly	Introduce Parameter Object or a small value class
Primitive Obsession	Overusing primitives instead of small domain types	Introduce a domain type (e.g., Money, EmailAddress)
Divergent Change	One class changes for many unrelated reasons	Split the class by responsibility (apply SRP)

KEY TAKEAWAY Code smells are not bugs — they're early warnings. Fixing them early keeps your code clean and future-ready.



REFACTORING TECHNIQUES

Detect complexity and code smells, then refactor safely — a 3-step process for cleaner, easier-to-change code.

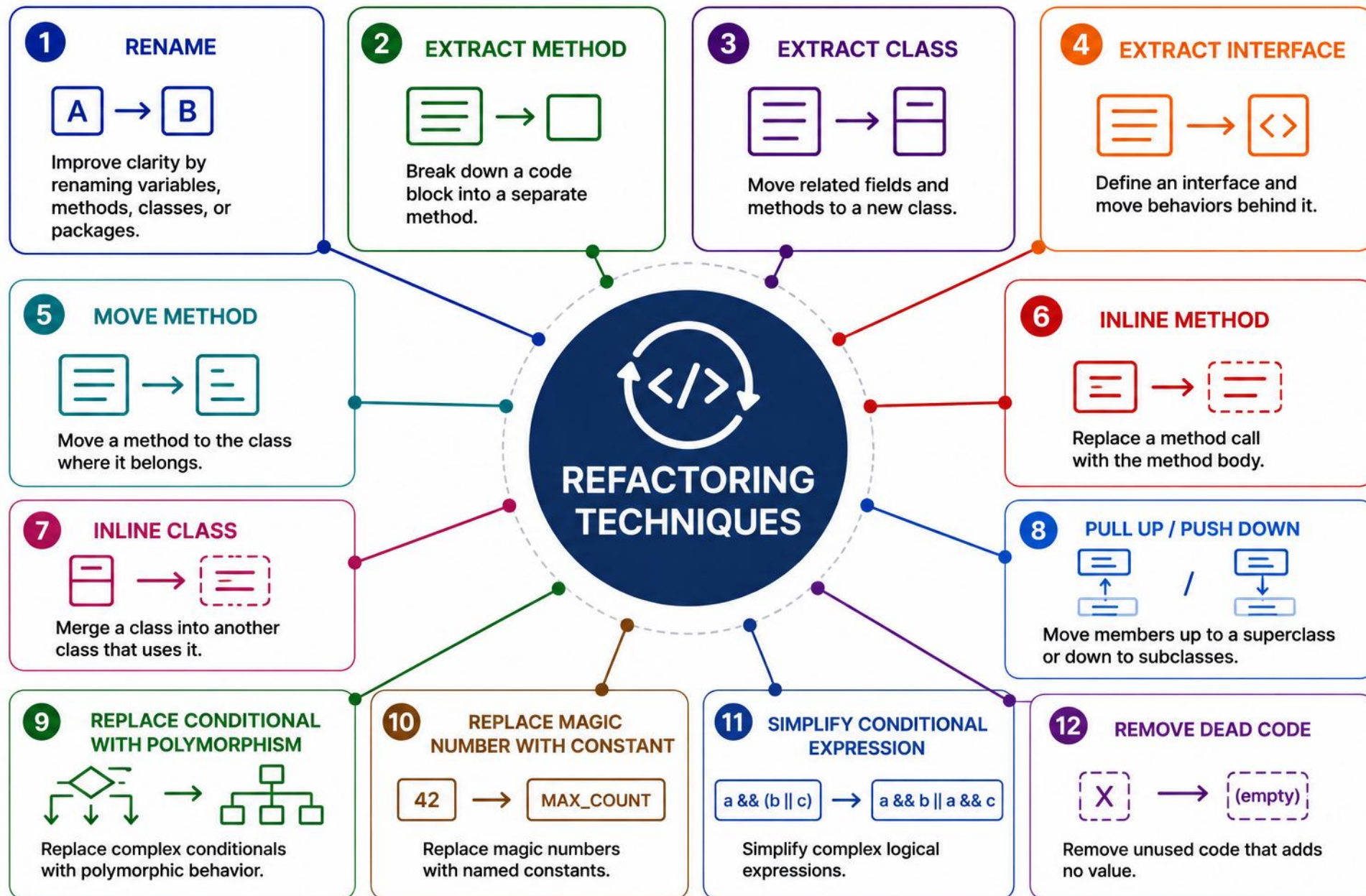
TECHNIQUE	WHAT IT DOES	BENEFIT
Extract Method	Break a long method into smaller, well-named methods	Improves readability and reusability
Extract Class	Move related fields/methods to a new class	Better cohesion and separation
Rename	Use meaningful names for variables, methods, classes	Easier to understand
Introduce Guard Clauses	Handle edge cases early, reduce nesting	Reduces complexity
Replace Conditional with Polymorphism	Replace type-checks with polymorphic behavior	Open for extension

SAFE REFACTORING PROCESS



- Detect complexity signals (deep nesting, high cyclomatic complexity, mixed abstraction levels); use polymorphism only for stable type variation. Before refactoring, establish a safety net -- tests where possible, or static analysis, version control, and small steps.

KEY TAKEAWAY Small, safe changes lead to big, long-term benefit -- always refactor behind a safety net, whether tests, analysis, or small incremental steps.



STATIC ANALYSIS ECOSYSTEM

Automated tools and CI/CD quality gates that analyze the whole codebase — separate from IntelliJ's live, in-editor feedback.

WHAT IT FINDS

- Bugs — e.g. null pointer dereference, resource leaks.
- Security vulnerabilities — e.g. SQL injection.
- Code smells — long methods, duplication.
- Coding standard violations and style issues.

POPULAR TOOLS (JAVA)

TOOL	PURPOSE
SonarQube	Centralized code-quality and security analysis across projects
Checkstyle	Source-style and convention checks
PMD	Source-based rules for potential defects and maintainability
SpotBugs	Bytecode analysis for likely defects

BEST PRACTICES

**Start Early**
Integrate SCA in IDE and pipeline.

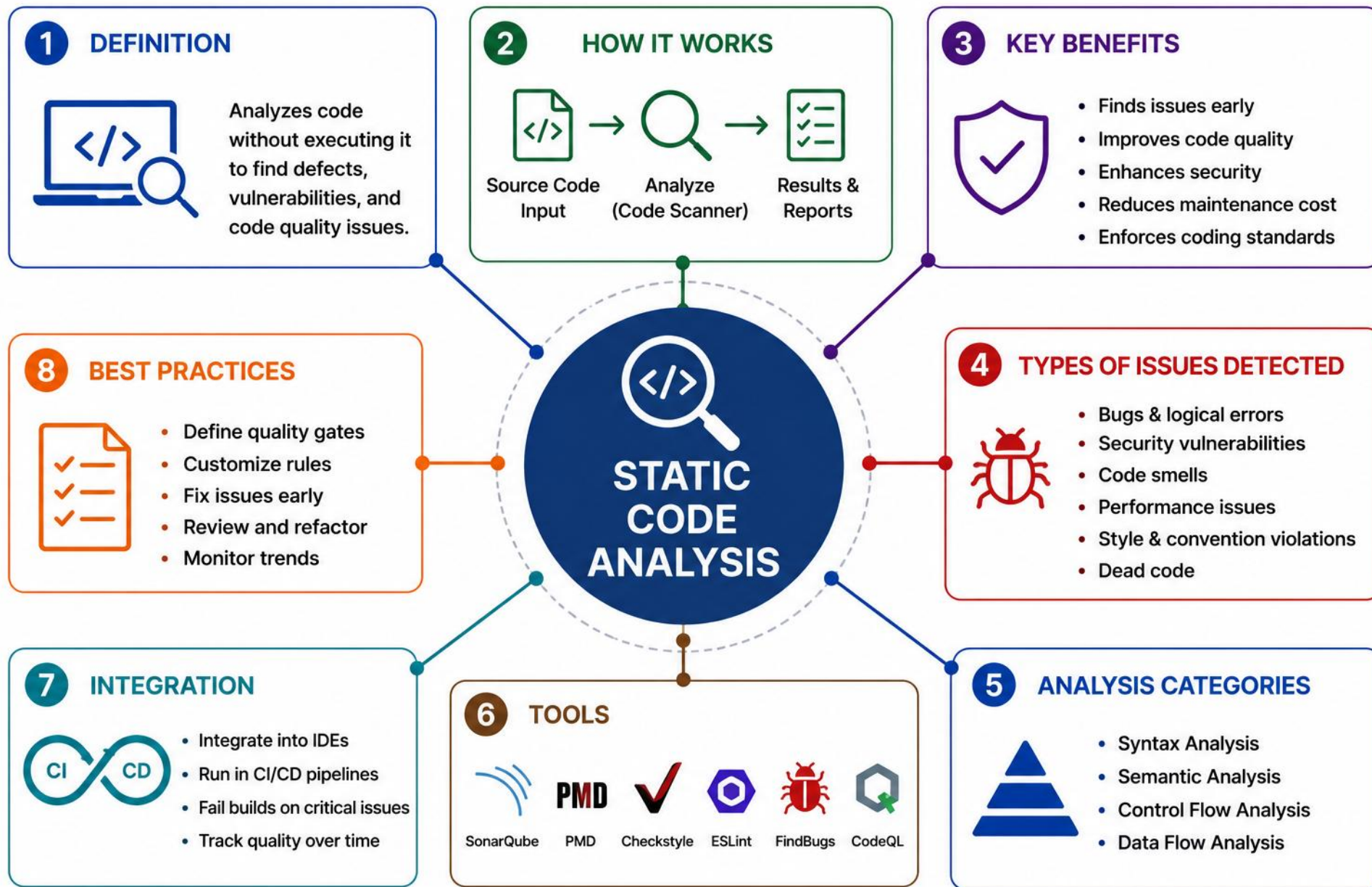
**Review Reports**
Act on issues, not just generate them.

**Measure Progress**
Track and reduce issues over time.

**Focus on Real Impact**
Not every warning is a bug.

Also part of SCA: syntax, semantic, control-flow & data-flow analysis, wired into CI/CD quality gates that fail builds on violations.

KEY TAKEAWAY Analyze early, fix early -- automated tools plus CI/CD gates catch what manual review misses.



INTELIJ CODE INSPECTIONS

IntelliJ IDEA gives local, in-editor feedback as you type -- quick fixes and a team-shared inspection profile, distinct from CI/CD-wide static analysis.

TYPES OF INSPECTIONS



Probable Bugs

Detects likely bugs and runtime issues.



Code Smells

Maintainability and readability problems.



Security Vulnerabilities

Insecure APIs, weak crypto, injection risks.



Formatting & Style

Enforces code style and formatting rules.



Performance Issues

Inefficient code and resource usage.



Duplicate Code

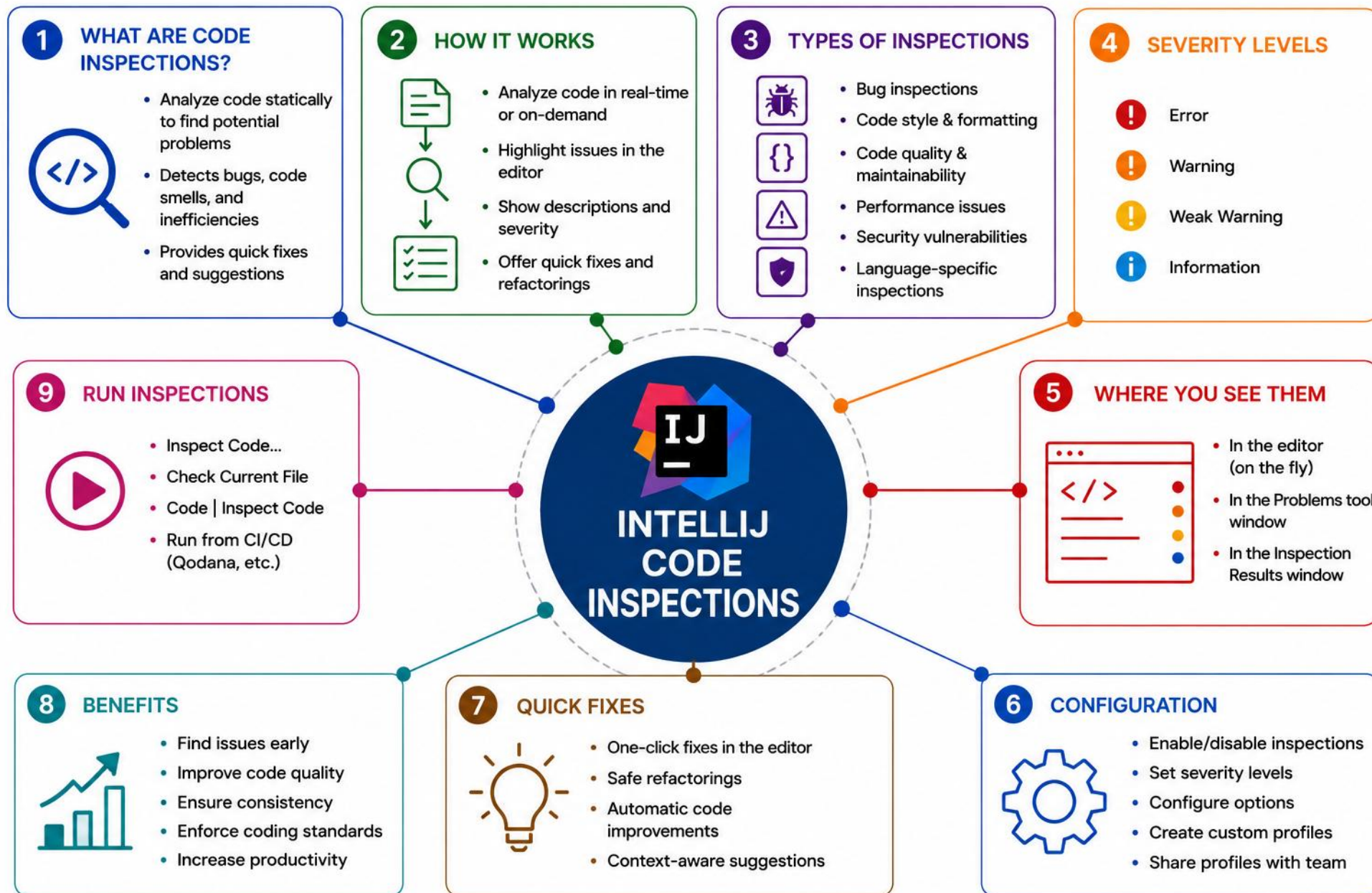
Copy-pasted or repeated logic.

USEFUL SHORTCUTS (WINDOWS/LINUX DEFAULTS)

- Inspect Code: Ctrl+Alt+Shift+I • Quick Fix: Alt+Enter • Reformat Code: Ctrl+Alt+L • Configure Inspections: Ctrl+Alt+S
- Shortcuts vary by operating system and keymap -- use Find Action (press Shift twice) to locate any command by name.

Severity levels: Error > Warning > Weak Warning > Info — configurable per rule under Settings → Editor → Inspections.

KEY TAKEAWAY Fix issues as IntelliJ flags them -- catching problems while you type is cheaper than catching them later in review or CI.



ENTERPRISE CODE REVIEW PRACTICES

Code reviews are a team quality gate — they improve design, reduce risk, and spread knowledge across the team.



WHAT TO REVIEW

- Correctness — does it do what it should, handle edge cases?
- Design & architecture — simple, extensible, well-structured?
- Readability, performance, security, and tests.

REVIEW PRINCIPLES

- Focus on the code, not the author — be constructive.
- Look for real risks, not style; automate style checks.
- Balance speed and quality — don't block progress needlessly.
- Keep pull requests small; explain intent and risk.
- Separate blockers from suggestions; track open comments.
- Review architecture and behavior, not personal preference.
- Require evidence for performance claims.

Backed by tooling: GitHub/GitLab PR workflows, CI/CD integration, and static analysis as quality gates.

KEY TAKEAWAY Great reviews create great code and great engineers — make code review a positive, consistent habit.



DEFENSIVE PROGRAMMING AT TRUST BOUNDARIES

Validate, normalize, and reject invalid data at the boundary — treat all external input as untrusted until proven otherwise.



Validate

- Check inputs at the boundary; reject null, blank, or out-of-range values immediately.



Normalize

- Convert to a consistent internal form (types, casing, trimming) before use.



Reject Invalid State

- Fail fast on invalid state; avoid ambiguous null returns -- use Optional or a documented exception where absence is exceptional.



Avoid Exposing Internal Details

- Don't leak internal representations, stack traces, or implementation details across the boundary.



Prefer Immutable Inputs

- Use final fields and immutable objects so validated state can't change unexpectedly.



Log Safely

- Log enough context to diagnose issues without recording sensitive data.

ANTI-PATTERNS TO AVOID

- Leaking internal exceptions or stack traces across the boundary; trusting caller-supplied data without checks; logging sensitive data; using Optional indiscriminately for every field, parameter, or return type.

KEY TAKEAWAY Defensive code fails loudly at the boundary -- catching bad data there is far cheaper than debugging it in production.

LAB OVERVIEW – CODING STANDARDS AND REFACTORING

Analyze non-compliant legacy code and refactor it to improve readability, maintainability, and design.

SCENARIO & STARTING QUALITY PROBLEMS

- You are maintaining a legacy CRM CustomerService that has grown inconsistent, duplicated, and poorly structured over time -- long methods, duplicated logic, and unclear names make it risky to change.
- Bring it in line with enterprise coding standards and refactor for long-term maintainability.

WORKFLOW

- Analyze the existing code and inspection results.
- Refactor using proven techniques, in small steps.
- Verify behavior with tests, then document changes.

CONSTRAINTS Prerequisites: Core Java, JUnit 5, IntelliJ IDEA, Maven. Time-boxed to the lab session -- deliverables include refactored code, tests, and a refactoring report.

KEY TAKEAWAY Good code is not written once — it is continuously improved. Refactor in small steps, commit often.

LAB TASKS AND DELIVERABLES

Apply coding standards and refactoring techniques to improve quality, readability, and maintainability.

#	TASK	DETAILS	FORMAT
1	Set Up the Project	Clone starter repo; import into IntelliJ IDEA.	n/a
2	Run Code Analysis	Inspect Code; review smells, complexity, violations.	Report
3	Review Coding Standards	Compare code against the provided standards doc.	n/a
4	Refactor the Code	Extract Method/Class, Remove Duplication, etc.	.java
5	Run & Fix Tests	Ensure all unit tests pass after refactoring.	Test results
6	Document Changes	Summarize issues found and improvements made.	.md/.docx

SUCCESS CRITERIA

- All critical/high inspection issues addressed; all unit tests pass; code follows the provided standards; refactoring report clearly summarizes the work.
- Each issue: fixed, suppressed with justification, or documented as not applicable -- don't change code just to satisfy a tool.
- Show behavior preserved: before/after test results, preserved public behaviors, before/after complexity evidence, and a commit per step.
- Estimated effort: 45 minutes; branch as feature/lab-refactor; commit frequently; open a PR before the deadline.

KEY TAKEAWAY Refactor incrementally, run tests often, and never change behavior — let tests be your safety net.

JAVA CODE QUALITY: DEFINITION OF DONE

A practical checklist for whether code is genuinely ready -- not just compiling, but meeting the module's standards.

- 1 Naming communicates intent.
- 2 Formatting is automated.
- 3 Methods and classes are cohesive.
- 4 Dependencies point in the correct direction.
- 5 Inputs are validated at boundaries.
- 6 Exceptions preserve useful context.
- 7 Duplication and unnecessary complexity are minimized.
- 8 Tests protect behavior.
- 9 Static-analysis findings are resolved or justified.
- 10 Code review is complete; documentation explains non-obvious decisions.

Bottom line: treat this checklist as your Definition of Done for any Java change -- not every box needs a tool, but every box needs an answer.

KEY TAKEAWAY Clean code is not written once -- it's a continuous practice. Use this checklist before you call any change done.

KNOWLEDGE CHECK (1 OF 2)

Test your understanding of coding standards, code smells, and refactoring techniques.

1. What's the primary goal of coding standards?

- A. Adding new features
- B. To reduce lines of code
- C. Consistency, readability, and maintainability**
- D. To eliminate all comments

3. What does refactoring mean?

- A. Adding new features
- B. Improving structure without changing behavior**
- C. Deleting unused code
- D. Rewriting from scratch

2. Which is an example of a code smell?

- A. Meaningful comments
- B. A class with 5000 lines of code**
- C. Proper use of constants
- D. Clear method names

4. A method has six related address parameters and is called from five locations. Which refactoring fits best?

- A. Extract Method
- B. Introduce Parameter Object**
- C. Rename
- D. Replace Magic Number

KEY TAKEAWAY Refactoring improves internal structure while preserving behavior — always keep tests as your safety net.

KNOWLEDGE CHECK (2 OF 2)

Four more questions, plus the full answer key.

5. A refactor changes the exception type and logging behavior while all tests still pass. Is it behavior-preserving?

- A. Yes -- passing tests always means behavior is preserved
- B. Not necessarily -- exceptions are part of behavior, too**
- C. Only if performance also improves
- D. No -- refactoring can never change any code

7. Why should methods be small and focused?

- A. They use fewer variables
- B. They're easier to test, understand, reuse**
- C. They compile faster
- D. They reduce class count

ANSWER KEY

• 1-C 2-B 3-B 4-B 5-B 6-B 7-B 8-B

6. What's a key benefit of code reviews?

- A. They replace automated tests
- B. They catch issues the original author may miss**
- C. They slow down development
- D. They're only for senior developers

8. What should you do BEFORE refactoring code?

- A. Remove all comments
- B. Ensure tests exist to verify behavior**
- C. Add more features
- D. Change all method names

KEY TAKEAWAY Write code that is easy to read, easy to change, and easy to trust — better code, better quality, better software.

MODULE 12 COMPLETE

Java Coding Standards and Best Practices

You can now write clean, consistent, maintainable Java code and apply proven refactoring techniques with confidence.